Java Looping Practice – 30 Beginner Problems (Definition, Approach, Code)

This document contains 30 beginner-level Java problems focused on looping statements (for, while). Each problem includes: 1. Definition – what the problem means 2. Approach – simple steps to solve 3. Example Java code Students should first try solving the problem themselves and then refer to the example solution.

1. Print Numbers from 1 to N

```
Definition:
Display numbers starting from 1 up to N.

Approach:
Use a for loop starting from 1 and ending at N.

Java Code:
for(int i=1;i<=n;i++){
    System.out.println(i);
}
```

2. Print Numbers from N to 1

```
Definition:
Print numbers in reverse order from N down to 1.

Approach:
Start loop at N and decrease until 1.

Java Code:
for(int i=n;i>=1;i--){
    System.out.println(i);
}
```

3. Print Even Numbers

```
Definition:
Even numbers are divisible by 2.

Approach:
Check each number from 1 to N and print if divisible by 2.

Java Code:
for(int i=1;i<=n;i++){
    if(i%2==0)
        System.out.println(i);
}
```

4. Print Odd Numbers

```
Definition:
Odd numbers are numbers not divisible by 2.

Approach:
Print numbers where remainder is not zero.

Java Code:
for(int i=1;i<=n;i++){
    if(i%2!=0)
```

```
        System.out.println(i);
}
```

5. Sum of First N Numbers

```
Definition:
Find the total of numbers from 1 to N.

Approach:
Use a loop and keep adding numbers to sum.

Java Code:
int sum=0;
for(int i=1;i<=n;i++){
    sum+=i;
}
System.out.println(sum);
```

6. Factorial of a Number

```
Definition:
Factorial multiplies all numbers from 1 to N.
Example: 5! = 120

Approach:
Multiply numbers from 1 to n.

Java Code:
int fact=1;
for(int i=1;i<=n;i++){
    fact*=i;
}
System.out.println(fact);
```

7. Multiplication Table

```
Definition:
Print multiplication table for a number.

Approach:
Loop from 1 to 10.

Java Code:
for(int i=1;i<=10;i++){
    System.out.println(num+" x "+i+" = "+(num*i));
}
```

8. Sum of Even Numbers

```
Definition:
Find the sum of all even numbers from 1 to N.

Approach:
Check if number is even and add.

Java Code:
int sum=0;
for(int i=1;i<=n;i++){
    if(i%2==0) sum+=i;
}
System.out.println(sum);
```

9. Sum of Odd Numbers

Definition:
Find sum of odd numbers between 1 and N.

Approach:
Add numbers where remainder is not zero.

Java Code:
```
int sum=0;
for(int i=1;i<=n;i++){
    if(i%2!=0) sum+=i;
}
System.out.println(sum);
```

10. Reverse a Number

Definition:
Reverse digits of a number.
Example: 1234 → 4321

Approach:
Use modulus and division.

Java Code:
```
int rev=0;
while(num>0){
    rev=rev*10+num%10;
    num/=10;
}
System.out.println(rev);
```

11. Count Digits

Definition:
Find how many digits exist in a number.

Approach:
Divide number by 10 until it becomes zero.

Java Code:
```
int count=0;
while(num>0){
    num/=10;
    count++;
}
System.out.println(count);
```

12. Sum of Digits

Definition:
Add all digits of a number.
Example: 123 → 6

Approach:
Extract digits using modulus.

Java Code:
```
int sum=0;
while(num>0){
    sum+=num%10;
    num/=10;
}
System.out.println(sum);
```

13. Product of Digits

Definition:
Multiply all digits of a number.

```
Approach:
Extract digits and multiply.

Java Code:
int product=1;
while(num>0){
    product*=num%10;
    num/=10;
}
System.out.println(product);
```

14. Palindrome Number

```
Definition:
Number that remains same when reversed.
Example: 121

Approach:
Reverse and compare.

Java Code:
int original=num;
int rev=0;

while(num>0){
    rev=rev*10+num%10;
    num/=10;
}

if(original==rev)
    System.out.println("Palindrome");
```

15. Prime Number

```
Definition:
A number divisible only by 1 and itself.

Approach:
Check divisibility from 2 to n/2.

Java Code:
boolean prime=true;
for(int i=2;i<=num/2;i++){
    if(num%i==0){
        prime=false;
        break;
    }
}
```

16. Fibonacci Series

```
Definition:
Each number is sum of previous two numbers.

Approach:
Use two variables.

Java Code:
int a=0,b=1;

for(int i=1;i<=n;i++){
    System.out.print(a+" ");
    int c=a+b;
    a=b;
    b=c;
}
```

17. Armstrong Number

```
Definition:
Sum of cube of digits equals number.
Example: 153

Java Code:
int sum=0,temp=num;

while(temp>0){
    int d=temp%10;
    sum+=d*d*d;
    temp/=10;
}
```

18. Strong Number

```
Definition:
Sum of factorial of digits equals number.
Example: 145

Java Code:
int sum=0,temp=num;

while(temp>0){
    int d=temp%10;
    int fact=1;

    for(int i=1;i<=d;i++)
        fact*=i;

    sum+=fact;
    temp/=10;
}
```

19. Largest Digit

```
Definition:
Find biggest digit in number.

Java Code:
int max=0;
while(num>0){
    int d=num%10;
    if(d>max) max=d;
    num/=10;
}
```

20. Smallest Digit

```
Definition:
Find smallest digit.

Java Code:
int min=9;
while(num>0){
    int d=num%10;
    if(d<min) min=d;
    num/=10;
}
```

21. Count Even Digits

```
Definition:
Count digits divisible by 2.
Java Code:
```

```
int count=0;

while(num>0){
    int d=num%10;
    if(d%2==0) count++;
    num/=10;
}
```

22. Count Odd Digits

```
Definition:
Count digits not divisible by 2.

Java Code:
int count=0;

while(num>0){
    int d=num%10;
    if(d%2!=0) count++;
    num/=10;
}
```

23. Perfect Number

```
Definition:
Sum of factors equals number.
Example: 6

Java Code:
int sum=0;

for(int i=1;i<num;i++){
    if(num%i==0)
        sum+=i;
}
```

24. Print Numbers Divisible by 3

```
Java Code:
for(int i=1;i<=n;i++){
    if(i%3==0)
        System.out.println(i);
}
```

25. Print Numbers Divisible by 5

```
Java Code:
for(int i=1;i<=n;i++){
    if(i%5==0)
        System.out.println(i);
}
```

26. Print Square of Numbers

```
Java Code:
for(int i=1;i<=n;i++){
    System.out.println(i*i);
}
```

27. Print Cube of Numbers

```
Java Code:
for(int i=1;i<=n;i++){
    System.out.println(i*i*i);
}
```

28. Star Triangle Pattern

```
Java Code:
for(int i=1;i<=5;i++){
    for(int j=1;j<=i;j++){
        System.out.print("* ");
    }
    System.out.println();
}
```

29. Reverse Star Pattern

```
Java Code:
for(int i=5;i>=1;i--){
    for(int j=1;j<=i;j++){
        System.out.print("* ");
    }
    System.out.println();
}
```

30. Power of a Number

```
Definition:
Calculate base^power.

Java Code:
int result=1;

for(int i=1;i<=power;i++){
    result*=base;
}

System.out.println(result);
```